

Acoustic Metamaterials — Planning Sheet

IB Group 4 Collaborative Sciences Project Date: 10 June 2026 | **Phase:** Planning | **Version:** 1.0

1. Project Framing

Central Question

How can acoustic metamaterials — engineered structures that manipulate sound waves — provide sustainable, futuristic alternatives to conventional architectural sound design?

Why This Topic Matters

- **Noise pollution** costs Europe ~€100 billion/year in health and productivity losses
- **Conventional soundproofing** (foam, mineral wool) is bulky, petroleum-derived, and performs poorly at low frequencies (traffic hum, HVAC drone)
- **Acoustic metamaterials** break the rules: they can absorb bass frequencies with structures 50× thinner than the wavelength, and can be made from waste plastic, agricultural byproducts, or biodegradable materials
- **Dual sustainability win:** tackles noise pollution AND material waste simultaneously

IB Criteria Alignment

- **Environmental impact:** life-cycle of materials, carbon footprint, circular economy
 - **Energy input/output:** passive operation (no power needed), embodied energy of fabrication
 - **Human benefit:** quieter living/working spaces, health improvements, sustainable cities (UN SDG 11)
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2. Scientific Foundations

Physics Domain (Lead: A, Deputy: B)

Wave Mechanics:

- Sound as pressure waves: wavelength $\lambda = c/f$ ($c \approx 343$ m/s in air)
- Low frequency = long wavelength. 100 Hz $\rightarrow \lambda = 3.43$ m. Traditional absorbers need $\lambda/4$ thickness (85 cm!). Metamaterials achieve this in 3 cm via path folding.
- Reflection, transmission, absorption: energy conservation $R + T + A = 1$

Resonance Physics:

- Helmholtz resonance: $f_0 = (c/2\pi) \times \sqrt{A / V \cdot L_{\text{eff}}}$ — air plug in neck oscillates against cavity spring
- Membrane resonance: $f_0 = (1/2\pi) \times \sqrt{k_{\text{eff}} / m_{\text{eff}}}$ — tensioned membrane + added mass
- Quarter-wave resonance: $f = c/(4L_{\text{eff}})$ — folded channels extend L_{eff} without increasing thickness

Metamaterial Mechanisms:

- **Local resonance** $\rightarrow$ band gaps at subwavelength scales

- **Impedance matching** → sound enters rather than reflects, then dissipates
- **Thermoviscous losses** → energy converted to heat in narrow channels
- **Bragg scattering** → periodic structure interference creates stop bands

Chemistry Domain (Lead: C, D)

Material Chemistry:

- **Polymers:** PET (polyethylene terephthalate), PVC (polyvinyl chloride) — upcycling waste into acoustic structures; polymer chain structure affects viscoelastic damping
- **Natural fibers:** cellulose/lignin matrix in bamboo, hemp, cork — porosity and fiber morphology control absorption
- **Bio-polymers:** alginate (from seaweed), chitosan — freeze-cast into porous foams; crosslinking density tunes mechanical properties

Chemical Analysis Methods:

- FTIR spectroscopy — verify polymer composition of recycled feedstock
- TGA/DSC — thermal stability, glass transition (critical for fire safety)
- Porosimetry — pore size distribution vs. acoustic performance
- Contact angle — hydrophobicity for durability in humid environments

Chemical Sustainability Metrics:

- Embodied carbon (kg CO₂-eq per kg material)
- Biodegradability / recyclability at end-of-life
- Toxicological profile (VOC emissions, health during fabrication)

Interdisciplinary Bridge

- **Materials selection** sits at the physics-chemistry boundary: physical properties (ρ , E , ν , η) determine acoustic behavior; chemical composition determines those physical properties
- **Fabrication processes** (3D printing, freeze-casting, compression molding) are governed by chemistry (rheology, curing, phase transitions) but produce geometries that determine acoustic physics

3. Project Direction: Chosen Approach

Primary Focus: Sustainable Metamaterial Panel for Low-Frequency Building Sound Absorption

Rationale:

1. Low-frequency noise (traffic, HVAC, industrial) is the hardest problem — and where metamaterials show the biggest advantage over conventional materials
2. Sustainability angle is strong: we can use recycled or bio-based materials
3. Buildable as a physical model: a panel with visible internal structure demonstrates the principle
4. Measurable: smartphone-based acoustic testing can quantify performance

Target Specification

Parameter	Target
Frequency range	200–1000 Hz (low-mid, where conventional foam fails)
Thickness	≤ 5 cm (demonstrate subwavelength advantage)
Absorption coefficient α	≥ 0.6 at target frequencies
Primary material	Recycled or bio-based (rPET, bamboo, cork, or cardboard)
Passive operation	No electronics, purely structural absorption

Design Concept: Hybrid Resonator Array + Porous Backing

Combine two mechanisms:

- 1. **Multiple tuned Helmholtz/membrane resonators** for targeted low-frequency peaks
- 2. **Porous absorption layer** (natural fiber felt) for broadband mid-frequency absorption

This hybrid approach is backed by 2025–2026 research showing that coherent weak-resonance coupling produces flatter, broader absorption curves than single mechanisms alone.

4. Workstreams & Task Allocation

Workstream A: Physics Design & Simulation

Lead: A | Support: B

Task	Description	Output
A1	Calculate target frequencies and required resonator dimensions using Helmholtz and quarter-wave formulas	Design parameters spreadsheet
A2	Design 3–5 resonator geometries tuned to staggered frequencies across 200–1000 Hz	CAD sketches / drawings
A3	Model acoustic performance using transfer matrix method (analytical, spreadsheet-based)	Predicted absorption curve
A4	Optimize resonator layout for mutual coupling (avoiding anti-resonance dips)	Final resonator array layout
A5	Design testing protocol: impedance tube dimensions, speaker/mic setup	Testing procedure document

Workstream B: Materials Chemistry

Lead: C | Support: D

Task	Description	Output
B1	Characterize available recycled/bio-based materials (composition, density, porosity)	Materials datasheet

Task	Description	Output
B2	Evaluate material suitability: damping properties, workability, sustainability metrics	Material recommendation report
B3	Prepare/test material samples for porous absorption layer	Tested samples with absorption data
B4	Investigate chemical treatments for durability (moisture resistance, fire retardancy)	Treatment protocol
B5	Life-cycle comparison: chosen material vs. conventional acoustic foam	LCA summary table

Workstream C: Physical Model Construction

Lead: B | Support: All

Task	Description	Output
C1	Source materials and tools (cardboard/PVC pipes/3D printing/wood)	Materials list with costs
C2	Build resonator units (3–5 units with different tuning)	Individual resonator prototypes
C3	Assemble full panel with integrated resonators + porous backing	Completed physical model
C4	Aesthetic finishing for presentation (cross-section visible, labeled)	Presentation-ready model

Workstream D: Testing & Measurement

Lead: A | Support: E

Task	Description	Output
D1	Set up measurement rig (impedance tube or speaker-box method with phone app)	Working measurement setup
D2	Measure absorption of individual resonators	Per-resonator frequency response
D3	Measure absorption of full panel	Overall absorption curve
D4	Compare experimental results with theoretical predictions	Validation report
D5	Iterate: identify gaps and propose design improvements	Improvement recommendations

Workstream E: Presentation & Documentation

Lead: E | Support: All

Task	Description	Output
E1	Research and write scientific background section	Background slides/content
E2	Document design process and decisions	Methods section
E3	Create graphs: predicted vs. measured absorption, LCA comparison	Data visualizations
E4	Design poster layout (graphs, photos, diagrams)	Poster draft
E5	Prepare digital presentation slides	Slide deck

5. Detailed Timeline

Phase 1: Planning & Research (June 10–11)

Date	Activity	Workstream
June 10 (Tue) 13:00–13:40	Introduction, science lecture, initial planning	All
June 10 (Tue) post-session	Distribute research: Physics principles (A,B), Chemistry/materials (C,D), Applications (E)	All
June 11 (Thu) 13:00–15:50	Consolidate research, finalize design concept, assign detailed tasks, begin calculations	All

Phase 2: Detailed Design & Preparation (June 12–15, independent work)

Deadline	Activity	Workstream
June 12 (Fri)	Complete resonator calculations (A1), complete materials characterization (B1)	A, C
June 13 (Sat)	Finalize CAD/sketches (A2), source materials (C1)	A, B, C
June 14 (Sun)	Complete simulations (A3, A4), prepare material samples (B3)	A, C
June 15 (Mon)	Design review: verify all calculations, confirm materials ready	All

Phase 3: Build & Test (June 16)

Date	Activity	Workstream
June 16 (Tue) 13:00–15:50	Build resonator units (C2), set up testing rig (D1), begin assembly	All
June 16 (Tue) 13:00–15:50	Initial testing, iterate design if needed (D2, D3, D5)	A, E

Phase 4: Analysis & Refinement (June 17–22, independent work)

Deadline	Activity	Workstream
June 17–18	Complete all measurements, data analysis (D2–D4)	A, E
June 19 (Fri)	Draft presentation structure, begin poster design (E1–E4)	E
June 20–21	Refine model, re-test if needed, finalize data visualizations	All
June 22 (Mon)	Full rehearsal: presentation walkthrough, poster review, model check	All

Phase 5: Final Presentation (June 23)

Time	Activity
8:00–11:40	Final preparations, printing, model touch-ups
13:00–14:20	Continued preparation
14:30–15:50	Final Presentation (Venue A105)
16:00–16:40	Reflection on ManageBac

6. Deliverables Checklist

Digital Presentation

- ☐ Title slide: project name, topic, team
- ☐ Scientific background: wave physics, metamaterial principles
- ☐ Sustainability framing: environmental problem + solution
- ☐ Design methodology: calculations, material selection rationale
- ☐ Results: predicted vs. measured absorption curves
- ☐ LCA comparison: our material vs. conventional alternatives
- ☐ Conclusion: what we achieved, what we'd improve
- ☐ References

Poster

- ☐ Eye-catching title and visual layout
- ☐ Key diagram: how acoustic metamaterials work (wave paths, resonance)
- ☐ Photo of physical model with labeled components
- ☐ Main graph: absorption coefficient vs. frequency (predicted + measured)
- ☐ Secondary graph: environmental impact comparison
- ☐ Brief bullet-point conclusions

Physical Model

- ☐ Visible internal structure (cross-section or transparent side)
- ☐ Labeled components (resonator cavities, necks, porous layer)
- ☐ Sturdy and portable
- ☐ Aesthetic presentation quality

7. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Resonator fabrication inaccurate → wrong tuning	Medium	High	Build extras, allow tuning adjustment (variable cavity volume)
Recycled material properties inconsistent	Medium	Medium	Characterize multiple samples, have backup commercial material
Measurement setup unreliable	Medium	High	Calibrate with known material, use multiple phone apps cross-checking
Model breaks during transport	Low	Medium	Reinforce structure, bring repair kit on presentation day
Not enough time for full testing	Medium	Medium	Prioritize 2–3 key frequencies if needed; simulation results are valid backup
Team member illness/absence	Low	High	All key knowledge documented; cross-train tasks across members

8. Resources Needed

Materials (estimate)

Item	Purpose	Est. Cost
Cardboard/foam board sheets	Main panel structure	~\$5
PVC pipes (various diameters) or 3D-printed parts	Helmholtz resonator bodies	~\$10
Natural fiber felt / recycled fabric	Porous absorption layer	~\$5
Wooden/base frame	Panel housing	~\$5
Hot glue, tape, fasteners	Assembly	~\$3
Total material cost estimate		~\$28

Equipment

- Smartphone with Phyphox or Spectroid app (free) — acoustic measurement
- Laptop with Audacity (free) — tone generation
- Speaker (Bluetooth or wired) — test signal source
- Ruler, utility knife, hot glue gun — construction
- Optional: 3D printer (school FabLab?) — precise resonator fabrication
- Optional: COMSOL Multiphysics (school license?) — advanced simulation

9. Key References

1. Comandini et al. (2025). "Architected acoustic metamaterials: An integrated design perspective." *Applied Physics Reviews*. — Comprehensive 53-page review of all physical mechanisms.
2. Ciaburro & Puyana-Romero (2025). "Sustainable Membrane-Based Acoustic Metamaterials Using Cork and Honeycomb Structures." *Buildings*, 14(9). — Fully sustainable, recyclable metamaterial.
3. Ciaburro & Puyana-Romero (2026). "Upcycled PVC-Based Metamaterials for Low-Frequency Sound Absorption." *Sustainability*, 18(5). — Waste PVC → acoustic panels.
4. *Building and Environment* (2026). "Blockfold origami ultrathin metamaterials." — 19 mm panel, $\alpha > 0.85$ from 440–1958 Hz.
5. *Nature Communications* (2025). "Deep-subwavelength acoustic-black-hole metamaterials." — 25–1200 Hz absorption, $\lambda/11$ thickness.
6. Liu et al. (2000). "Locally Resonant Sonic Materials." *Science*, 289. — The founding paper of acoustic metamaterials.
7. IB Collaborative Sciences Project Guide — Official assessment criteria and requirements.

10. Appendix: Quick Reference Formulas

Speed of sound:	$c = 331.3 \times \sqrt{1 + T/273.15}$ [m/s, T in °C] $c \approx 343$ m/s at 20°C
Wavelength:	$\lambda = c / f$
Helmholtz f_0 :	$f_0 = (c/2\pi) \times \sqrt{A / V \cdot L_{\text{eff}}}$ $L_{\text{eff}} = L_{\text{neck}} + 0.85 \times d$ (flanged) or $0.7 \times d$ (unflanged)
Quarter-wave tube:	$f_0 = c / (4L)$
Sound absorption:	$\alpha = 1 - R ^2$ (where R = reflection coefficient) Measured via impedance tube (ISO 10534-2)
Transmission loss:	$STL = 10 \times \log_{10}(1/\tau)$ [dB] $\tau = \text{transmitted} / \text{incident power}$